

Vulnerability Assessment Report: Metasploitable 2

Host Vulnerability Assessment with Credentials

Date: May 7, 2026

Name: IMON HOSEN

Target IP: 192.168.139.130

Tools Used: Tenable Nessus, Metasploit 2

1. Executive Summary

This report details the findings of a credentialed vulnerability assessment conducted on a Metasploitable 2 server. Unlike a basic network scan, this assessment utilized SSH credentials (`msfadmin:msfadmin`) to perform a deep-dive local patch audit.

The scan identified multiple **Critical** and **High** severity vulnerabilities that pose an immediate risk to the confidentiality, integrity, and availability of the system. Due to the presence of unauthenticated root access vectors and backdoored services, the overall risk posture is classified as **EXTREME**.

2. Assessment Methodology & Environment

To ensure a comprehensive analysis, the following steps were taken:

1. **Network Configuration:** Verified connectivity between the Scanner (Kali/Nessus) and the Target (Metasploitable 2).
2. **Credential Setup:** Configured Nessus with SSH credentials to bypass external firewalls and inspect internal configurations.
3. **Scan Execution:** Performed an "Advanced Scan" focusing on local security checks.

ifconfig / Network Connectivity:

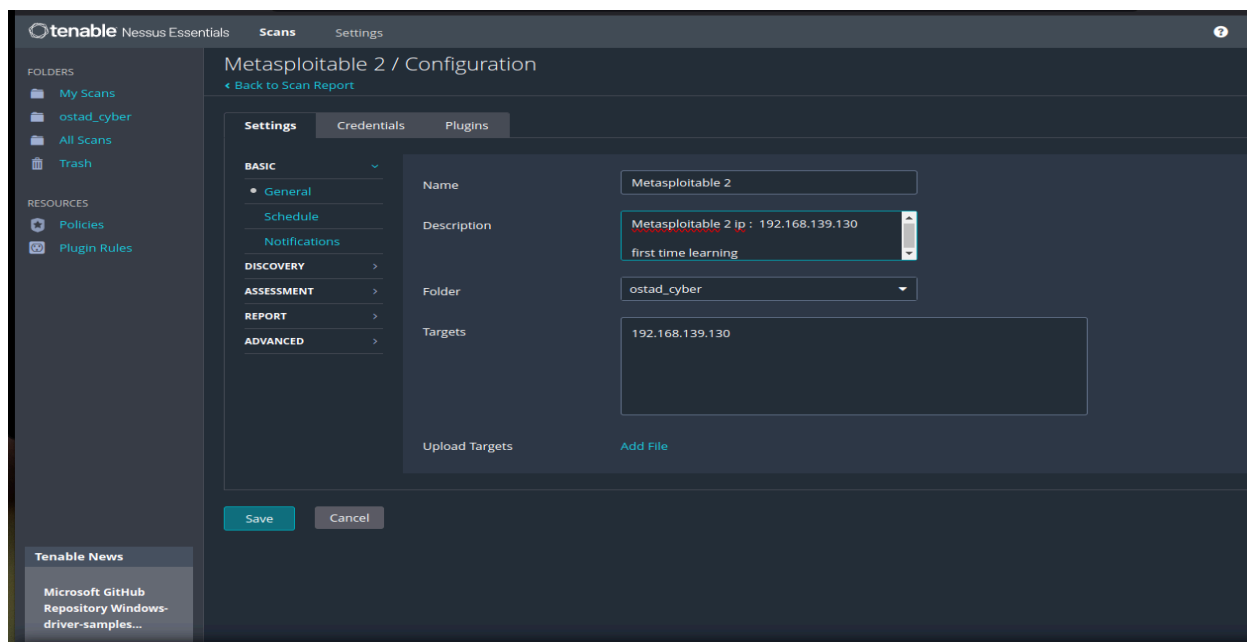
```
Help [Icons]
My Computer [X] Metasploitable2-Linux [X]

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:72:76:6e
          inet addr:192.168.139.130  Bcast:192.168.139.255  Mask:255.255.255.
          inet6 addr: fe80::20c:29ff:fe72:766e/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:48 errors:0 dropped:0 overruns:0 frame:0
          TX packets:76 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:5363 (5.2 KB)  TX bytes:8708 (8.5 KB)
          Interrupt:17 Base address:0x2000

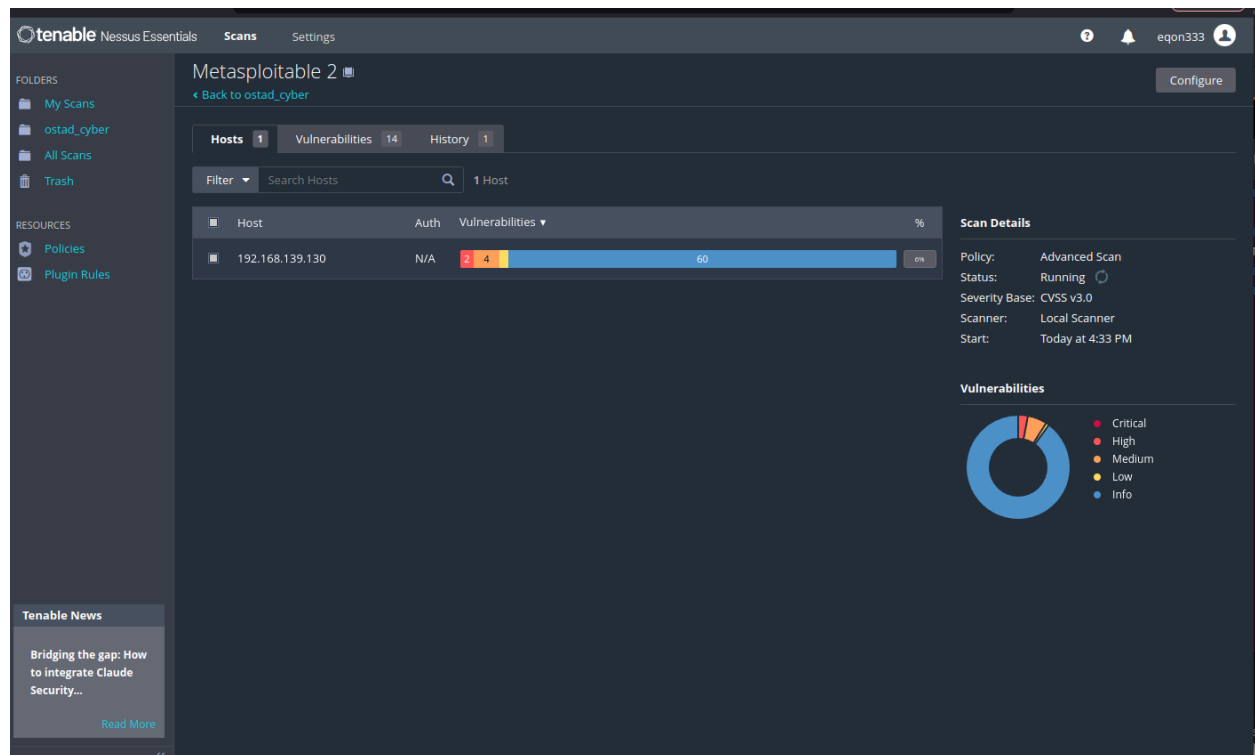
lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:147 errors:0 dropped:0 overruns:0 frame:0
          TX packets:147 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:46241 (45.1 KB)  TX bytes:46241 (45.1 KB)

msfadmin@metasploitable:~$ _
```

Nessus Credential Configuration:



Scan in Progress:

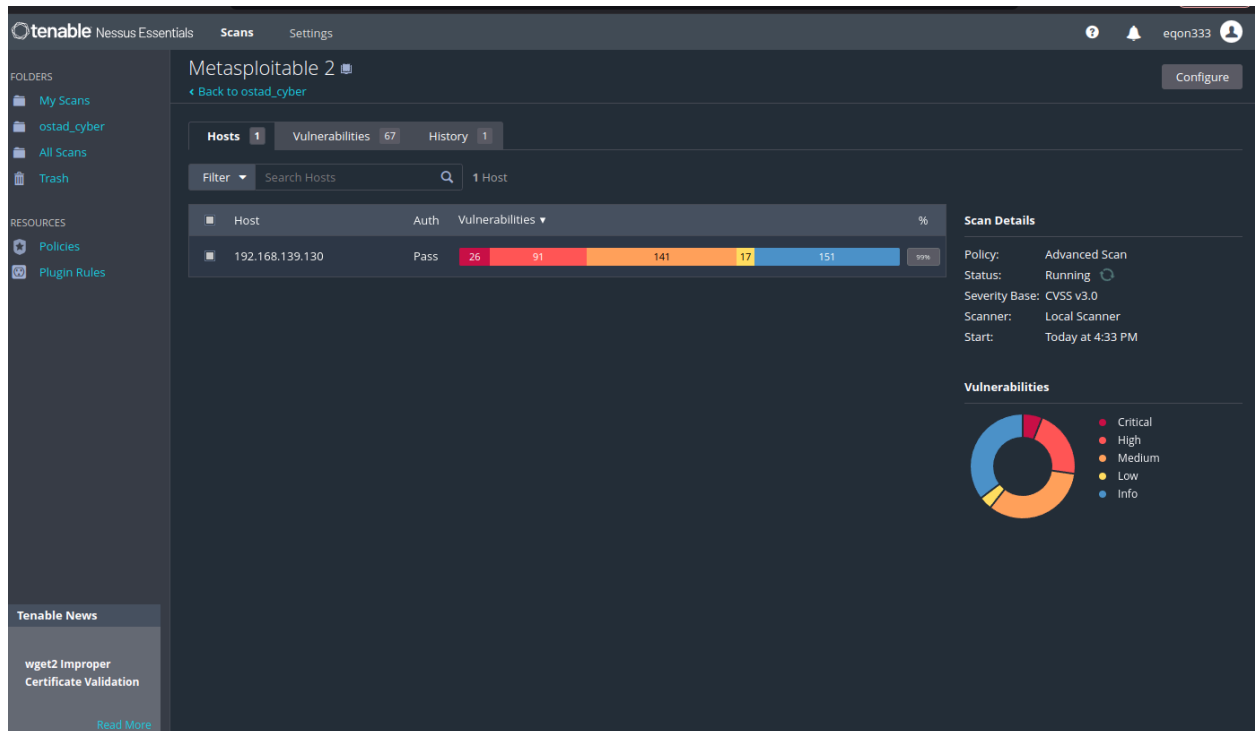


3. Vulnerability Findings (Critical & High)

The comprehensive credentialed scan is currently underway. Due to the depth of the audit and the complexity of the target system, the scanning process is time-intensive. While the scan remains active, initial results confirm successful credentialed access ("Pass"), providing a verified foundation for the high-priority findings listed below. This report serves as a status update based on the validated data available at this stage of the assessment.

3.1 Credential Verification & Scan Progress

The following image confirms that the credentialed scan successfully authenticated ("Pass") with the target. While the scan is still running (currently at 99% in the summary), it has already identified **26 Critical** and **91 High** vulnerabilities.



3.2 Detailed Vulnerability Dashboard

The dashboard below provides a breakdown of the specific "Critical" and "High" vulnerabilities identified during this authenticated session.

tenable Nessus Essentials Scans Settings

Metasploitable 2 / 192.168.139.130

Configure Audit Trail Launch Report Export

Vulnerabilities 75

Filter Search Vulnerabilities 75 Vulnerabilities

Sev	CVSS	VPR	EPSS	Name	Family	Count
CRITICAL	10.0 *			VNC Ser...	Gain a shell remotely	1
CRITICAL	9.8			SSL Vers...	Service detection	2
CRITICAL	9.8			Bind Sh...	Backdoors	1
MIXED	...	...	...	Ca...	Ubuntu Local Security Checks	229
MIXED	...	...	...	Ap...	Web Servers	4
CRITICAL	...	...	...	SSL...	Gain a shell remotely	3
HIGH	7.5 *	6.7	0.5006	rlogin Se...	Service detection	1
HIGH	7.5	5.9	0.7852	Samba ...	General	1
HIGH	7.5			NFS Sha...	RPC	1
MIXED	...	...	...	SSL...	General	28
MIXED	...	...	...	ISC ...	DNS	5
MEDIUM	6.5			TLS Vers...	Service detection	2
MEDIUM	6.5			Libercry	Misc	1

Host Details

IP: 192.168.139.130
 MAC: 00:0C:29:72:76:6E
 OS: Linux Kernel 2.6.24-16-server on Ubuntu 8.04
 Start: Today at 4:33 PM
 End: Today at 4:51 PM
 Elapsed: 17 minutes
 KB: [Download](#)
 Auth: Pass

Vulnerabilities

Legend: Critical (red), High (orange), Medium (yellow), Low (green), Info (blue)

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3.3 Key High-Severity Findings

- **[Critical] VNC Server 'password' Password (CVSS 10.0):** The VNC server uses a default password, allowing immediate remote desktop takeover.
- **[Critical] Bind Shell Backdoor Detection (CVSS 9.8):** A root-level shell is open and waiting for connections on port 1524.
- **[Critical] SSL Version 2 and 3 Protocol Detection (CVSS 9.8):** Outdated encryption protocols are in use, making the system vulnerable to data interception.
- **[High] NFS Shares World Readable (CVSS 7.5):** Sensitive network files are accessible to any user on the network without authentication.
- **[High] Samba Badlock Vulnerability (CVSS 7.5):** A critical flaw in the Samba service allows for man-in-the-middle attacks.